THEORY REPORT

Assignment 7

Title: Analyze Wi-Fi Communication to Identify Insecure Network Traffic and Recommend Suitable Security Measures

1. Aim

To analyze Wi-Fi communication, identify insecure network traffic and protocols, and recommend suitable security measures to protect data from interception and network-based attacks.

2. Objectives

1. To understand the working of Wi-Fi communication.
2. To study network packets transmitted over a Wi-Fi connection.
3. To identify insecure or unencrypted network traffic.
4. To understand risks associated with open and weakly secured Wi-Fi networks.
5. To recommend appropriate security measures for protecting Wi-Fi communication.

3. Introduction

Wi-Fi is a wireless communication technology that allows devices to connect to a network without physical cables. It is widely used in homes, educational institutions, offices, and public places. Since communication takes place through radio signals, attackers within an appropriate range may attempt to observe or manipulate network traffic.

Network packet analysis tools such as **Wireshark** can be used in an authorized environment to examine communication between devices. By analyzing protocols, packet information, and encryption characteristics, it is possible to identify insecure traffic such as unencrypted HTTP communication, exposed DNS queries, and suspicious or unexpected network activity.

4. Wi-Fi Communication

A typical Wi-Fi communication process involves:

Client Device → Wi-Fi Access Point → Router/Network → Internet

The client device communicates with the access point using wireless protocols. The access point forwards network traffic to the appropriate destination.

Wi-Fi security commonly uses mechanisms such as **WPA2** and **WPA3** to protect wireless communication. Older or open configurations provide weaker protection and may expose users to greater risks.

5. Network Traffic Analysis

Wi-Fi traffic can contain different protocols, each serving a specific purpose.

Protocol	Purpose	Security Concern
ARP	Maps IP addresses to MAC addresses	Can be abused for ARP spoofing
DNS	Resolves domain names to IP addresses	Queries may reveal browsing activity
HTTP	Transfers web data	Data can be transmitted without encryption
HTTPS	Secure web communication	Provides encryption through TLS
TCP	Reliable transport of data	Security depends on the application protocol
UDP	Connectionless communication	Security depends on the application protocol

6. Identification of Insecure Traffic

During an authorized packet analysis, the following indicators can be examined:

1. Unencrypted HTTP traffic:
HTTP does not provide encryption at the application layer. Sensitive information transmitted through HTTP may therefore be exposed to network observers.

2. Open Wi-Fi network:
An open wireless network does not require Wi-Fi authentication encryption such as WPA2 or WPA3. This increases the possibility of traffic observation and other attacks.

3. Unprotected or suspicious DNS communication:
Traditional DNS queries may reveal the domain names being requested. Secure DNS mechanisms can provide stronger privacy depending on the configuration.

4. ARP-related anomalies:
Unexpected or inconsistent ARP behavior may indicate ARP spoofing or other local-network attacks and should be investigated.

7. Security Risks

Insecure Wi-Fi communication may expose users to:

- Eavesdropping
- Man-in-the-Middle (MITM) attacks
- ARP spoofing
- Session interception
- Credential exposure
- Browsing-activity disclosure
- Data modification

- Unauthorized network access

The severity of a risk depends on the network configuration, protocols being used, and whether application-layer encryption is present.

8. Recommended Security Measures

The following measures can improve Wi-Fi security:

1. **Use WPA2 or WPA3:** Configure wireless networks with strong modern encryption rather than open or obsolete security mechanisms.
2. **Use HTTPS:** Websites and applications should use HTTPS/TLS to protect data at the application layer.
3. **Avoid open Wi-Fi for sensitive activities:** Do not perform banking or other highly sensitive activities on untrusted networks unless appropriate additional protection is used.
4. **Use a VPN on untrusted networks:** A reputable VPN can provide an additional encrypted tunnel for network traffic.
5. **Use strong Wi-Fi passwords:** Avoid easily guessable passwords.
6. **Keep router firmware updated:** Security updates can address known vulnerabilities.
7. **Disable unnecessary features:** Unneeded wireless administration and legacy features should be disabled.
8. **Monitor network activity:** Unexpected devices or abnormal traffic should be investigated.
9. **Use secure DNS where appropriate:** Encrypted DNS can reduce exposure of DNS queries to network observers.
10. **Separate guest devices:** Guest networks can help isolate untrusted devices from internal resources.

9. Observation

During Wi-Fi traffic analysis, protocols such as ARP, DNS, TCP, and HTTP/HTTPS can be observed. Unencrypted HTTP communication represents a significant security concern because application data may be transmitted without encryption. Open or weakly secured wireless networks also provide less protection against local traffic interception.

10. Result

Wi-Fi communication was analyzed based on network protocols and encryption characteristics. Insecure traffic such as unencrypted communication and potentially exposed network information was identified, and appropriate security measures were recommended to reduce the possibility of interception and network attacks.

11. Conclusion

Wi-Fi communication must be protected at both the **wireless-network layer** and the **application layer**. Using WPA2/WPA3 protects the wireless connection, while HTTPS/TLS protects application data during transmission. Packet analysis helps identify insecure protocols and abnormal communication patterns. Therefore, strong Wi-Fi encryption, secure application protocols, updated network equipment, and cautious use of public networks are essential for maintaining secure wireless communication.

